

Student Name: TODO

Collaboration Statement:

Total hours spent: TODO

I consulted the following human, textbook, or AI resources:

- TODO
- TODO
- ...

By turning this document in, I attest that I have followed the [CS 136 Collaboration Policy]. All solutions represent my own work. No solution text in this document was directly provided by other humans or artificial agents.

Contents

1: Solution	2
2a: Solution	3
2b: Solution	4
2c: Solution	5
3a: Solution	6
3b: Solution	7
3c: Solution	8
4: Solution	9

Problem 1

Suppose 10% of a given population has COVID-19 and they all test. The COVID-19 antigen test has 80% accuracy when taken by someone with COVID-19 (i.e. 80% chance of coming back positive) and is 99% accurate when taken by someone without COVID-19 (i.e. 99% chance of coming back negative). If you get a positive antigen test result, what is the probability you have COVID-19?

1: Solution

TODO YOUR SOLUTION HERE.

2: Problem Statement

Suppose you flip a fair coin 10 times (probability of heads is 0.5).

Let $X_i = 1$ if the i th flip **and** the $i + 1$ th flip were heads, and 0 otherwise (for $i = 1, \dots, 9$). Thus each X_i is tracking whether a "back-to-back heads" happened at a particular point in the 10-flip sequence.

2a: Problem Statement

What is $E[X_i]$, the expected value of X_i ?

2a: Solution

TODO YOUR SOLUTION HERE

2b: Problem Statement

Are X_i and X_{i+1} independent? Show your answer mathematically. *Note: the answer does not depend on the value of i .*

2b: Solution

TODO YOUR SOLUTION HERE

2c: Problem Statement

Let $X = \sum_{i=1}^9 X_i$. We could describe X as the “total number of back-to-back heads observed” (although this could be a slightly confusing description, because 3 heads in a row are being counted as 2 different back-to-back heads by this equation).

2c: Solution

TODO YOUR SOLUTION HERE

3: Problem Statement

Let X, Y, C be binary random variables (i.e. they can each take a value of 0 or 1). Consider the following joint distribution:

x	y	c	$p(x, y, c)$
0	0	0	0.15
0	1	0	0.1
1	0	0	0.15
1	1	0	0.1
0	0	1	0.1
0	1	1	0.05
1	0	1	0.2
1	1	1	0.15

3a: Problem Statement

What is $P(C = 1)$?

3a: Solution

TODO YOUR SOLUTION HERE

3b: Problem Statement

What is $P(C = 0|Y = 0)$?

3b: Solution

TODO YOUR SOLUTION HERE

3c: Problem Statement

Which of the following statements is true? You don't need to provide math showing the other 3 are false; just show your work for the true one.

- X and Y are independent.
- X and Y are conditionally independent given C .
- X and Y are conditionally independent given $C = 0$.
- X and Y are conditionally independent given $C = 1$.

3c: Solution

TODO YOUR SOLUTION HERE

4: Problem Statement

Let X be the result of a fair, six-sided die roll, so X has equal probability of taking any integer value 1 to 6.

What is the variance $Var(X)$ of X ?

4: Solution

TODD YOUR SOLUTION HERE